Route D: Boussinesq solitons from the quantum Toda chain

Project *quantum_boussinesq_solitons*, Phase 2b

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Abstract

We carry out the first steps of the top-ranked route of the Phase 2 report. The quantum Toda chain is exactly solvable, and its small-amplitude long-wave limit is the bad Boussinesq equation. We find:

1. In the classical limit, the asymptotic Bethe ansatz reproduces the Toda soliton sector *exactly*: energies, momenta, two-soliton position shifts, and the ordering-dependent "topological" cross terms of Phase 2.
2. The label-frame semiclassical phase of Toda solitons has the closed form

$$\delta_T = (p_1 - p_2)\log A_T - (\Pi_1 M_2 - \Pi_2 M_1),$$

   with p the Lax eigenvalue.
3. In the Boussinesq scaling limit, δ_T converges to $3b$ times the Phase 2 Boussinesq phase, *including its topological term*.
4. At finite $\hbar$ we compute the $O(\hbar)$ quantum corrections.

We also identify a structural limitation: Toda reaches only the small-amplitude corner of the Boussinesq soliton sector.

1 Setting

The Toda chain is

$$H = \sum_n \left(\tfrac{1}{2}p_n^2 + e^{-r_n}\right), \qquad r_n = x_{n+1} - x_n,$$

with mean spacing a and $b = e^{-a/2}$.

Classical solitons. The one-soliton is $e^{-r_n} = b^2[1 + \sinh^2\kappa\,\mathrm{sech}^2(\kappa n - s\,b\,t\sinh\kappa)]$, with $s = \pm 1$ the direction. Computing its charges numerically on the lattice gives

$$M = \sum(r_n - a) = -2\kappa, \qquad \Pi = \sum p_n = 2sb\sinh\kappa, \qquad E = 2b^2(\sinh\kappa\cosh\kappa - \kappa).$$

(c) `scripts/toda_soliton_classical.py`
The pull-back of $\Omega = \sum dp_n \wedge dx_n$ to the family (κ, n_0) is

$$dP_n \wedge dn_0, \qquad P_n = 4sb(\kappa\cosh\kappa - \sinh\kappa),$$

and dE/dP_n equals the label velocity $b\sinh\kappa/\kappa$. **(c)** `scripts/toda_label_symplectic.py` The Eulerian total momentum Π is *not* canonically conjugate to the soliton position.

Quantum two-body problem. The relative Hamiltonian q^2+e^{-x} is solved by $K_{2iq/\hbar}(2e^{-x/2}/\hbar)$. Its reflection amplitude is

$$R = \frac{\Gamma(i\kappa)}{\Gamma(-i\kappa)}\,\hbar^{2i\kappa}, \qquad \kappa = \frac{\Delta p}{\hbar}.$$

(c) (derived here; standard, but not compared line by line with [1, 2]). The asymptotic Bethe equations are $p_j L + \sum_l \Theta(p_j - p_l) = 2\pi\hbar I_j$ with

$$\Theta(D) = 2\hbar\,\mathrm{Im}\log\Gamma(1 + iD/\hbar) + 2D\log\hbar.$$

This has the classical limit $2(D\log|D| - D)$. The method is that of Sutherland and Mertens **(b)** [3, 4].

2 Classical limit of the Bethe ansatz

Proposition 1 (**(c)**, `scripts/toda_bethe_classical.py`)**.** *As $\hbar \to 0$ the ground-state quasi-momenta fill $[-w, w]$ with $w = 2b$, with the arcsine density of total weight $1/a$. This is the logarithmic equilibrium measure, which equals the Lax spectrum of the static lattice. It reproduces the ground-state energy per particle b^2.*

A particle excitation at $p = 2sb\cosh\kappa$, the soliton's Lax eigenvalue, has:

- *dressed momentum exactly P_n/a, the physical-frame canonical momentum;*
- *dressed energy, relative to the edge excitation, exactly the soliton energy E.*

Both hold to 10^{-8}; the Chebyshev method is exact up to truncation.

Proposition 2 (**(c)**, `scripts/toda_dressed_phase_check.py`, `scripts/toda_ba_vs_label.py`)**.** *Let $\Phi(1|2)$ be the dressed Bethe phase of excitation 1 in the presence of 2.*

1. *The exact classical physical-frame shifts of the two-soliton solution are*

   $$\Delta x_1 = a\Delta n_1 + M_2, \qquad \Delta x_2 = a\Delta n_2 - M_1,$$

   the extra terms being the displacement caused by the other soliton's compression. They are given by

   $$\Delta x_2 = -\partial\Phi(1|2)/\partial P_2, \qquad \Delta x_1 = +\partial\Phi(2|1)/\partial P_1.$$

2. *$\Phi(1|2) + \Phi(2|1)$ is* not *separable. Its mixed derivative equals exactly*

   $$M_1'(P_{n2}'/a - \Pi_2') + M_2'(P_{n1}'/a - \Pi_1').$$

   That is, the Bethe ansatz carries precisely the topological cross terms of the Phase 2 mechanism.

3. *Specifically,*

   $$\Phi(1|2) = \delta_T + M_1(P_{n2}/a - \Pi_2) + f(\kappa_1), \qquad \Phi(2|1) = -\delta_T + M_2(P_{n1}/a - \Pi_1) + g(\kappa_2),$$

   to machine precision.

3 Label frame and the Boussinesq limit

Proposition 3 (**(c)**, `scripts/toda_label_symplectic.py`, `scripts/toda_scaling_fast.py`). *1. On the lattice, two-soliton scattering satisfies*

$$\sum dP_{n,j} \wedge d\Delta n_j = d\Pi_2 \wedge dM_1 - d\Pi_1 \wedge dM_2$$

to 30 digits. This is the exact analogue of the Boussinesq identity.

2. The Darboux phase is

$$\boxed{\delta_T = (p_1 - p_2)\log A_T + M_1 M_2 (v_1 - v_2) = (p_1 - p_2)\log A_T - (\Pi_1 M_2 - \Pi_2 M_1)}$$

with $p = 2sb\cosh\kappa$ and v the label velocity. The ratio of the remainder to $M_1 M_2(v_1 - v_2)$ is 1 to ten digits.

3. With $\kappa_j = \sqrt{3}\beta_j \to 0$ and the action scale $S_{\mathrm{Toda}} = 3bS_{\mathrm{Bouss}}$,

$$\delta_T/(3b\,\delta_{\mathrm{bad}}) \to 1$$

for overtaking and *head-on collisions. The error is 1×10^{-5} at $\beta \sim 3 \times 10^{-3}$.*

At leading order the head-on Boussinesq phase is $4\beta_1\beta_2$. Of this, $\frac{16}{3}\beta_1\beta_2$ comes from the topological term and $-\frac{4}{3}\beta_1\beta_2$ from the dynamical term. Agreement in the head-on case is therefore a direct confirmation of the topological term, by an exactly solvable quantum model.

Remark 1 (**(d)**). Toda and Boussinesq share the structure

$$\delta = (\text{difference of spectral parameters}) \times \log A - c\,(\Pi_1 M_2 - \Pi_2 M_1),$$

with $c = 1$ (Toda) and $c = \frac{2}{3}$ (Boussinesq, in its normalisation). The split between the two terms is not invariant under the scaling limit; only the sum is. We conjecture the structure is universal for integrable models with continuous kink charge (KdV included) and suggest testing it on KdV.

4 Finite $\hbar$: quantum corrections

We solved the finite-$\hbar$ Bethe equations (Nyström method, node spacing $< \hbar/8$) for $\hbar = 0.1, 0.05, 0.025$ at $a = 1$ (`scripts/toda_bethe_quantum.py`, `scripts/toda_quantum_corrections.py`). All quantities converge linearly in $\hbar$ to the classical ones. Richardson extrapolation gives the one-loop soliton dispersion correction at fixed physical momentum, $E(P) = E_{\mathrm{cl}}(P) + \hbar E_1(P)$:

P	E_{cl}	E_1
0.60	0.39347	+0.0301
0.40	0.25768	+0.0203

(c) (numerical)

The $O(\hbar)$ phase corrections Φ_1 also converge. However, they contain convention-dependent constants: $\pm\pi/2$ from the Stirling branch of Θ, and quantum-number shifts. We have not yet separated out a convention-free quantity. **(c)**/**(d)**

5 The limitation, and what would give an exact Boussinesq S-matrix

The Toda continuum limit produces the bad Boussinesq equation only at *leading order in amplitude*. Toda solitons therefore test the Boussinesq phase only for $\beta \to 0$, where Boussinesq solitons are KdV-like with weak head-on coupling. Finite-β Boussinesq solitons have widths comparable to the dispersion length, which is where the bad equation is ill-posed. On the lattice this is the lattice spacing, where Toda and Boussinesq differ at $O(1)$. **(c)**

Consequently:

- In the double-scaling limit $\beta \to 0$, $\hbar \propto \beta^2$ (so that $\delta/\hbar = O(1)$), the quantum Toda dressed S-matrix gives an *exact quantum* S-matrix for small-amplitude Boussinesq solitons. In Bethe-ansatz terms this is an edge-scaling problem: excitations within $O(\hbar)$ of the Fermi edge, with the digamma kernel. It is well posed and is the next computation. It would extend Pustilnik–Matveev's analysis of the KdV crossover [5] to bidirectional (head-on) scattering. **(d)**

- An exact S-matrix for finite-amplitude Boussinesq solitons requires a quantum regularisation that agrees with Boussinesq at the dispersion scale. We know of no integrable lattice with that property. The ill-posedness of the bad equation is precisely the statement that the continuum theory alone does not provide one. **(c)/(d)**

References

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